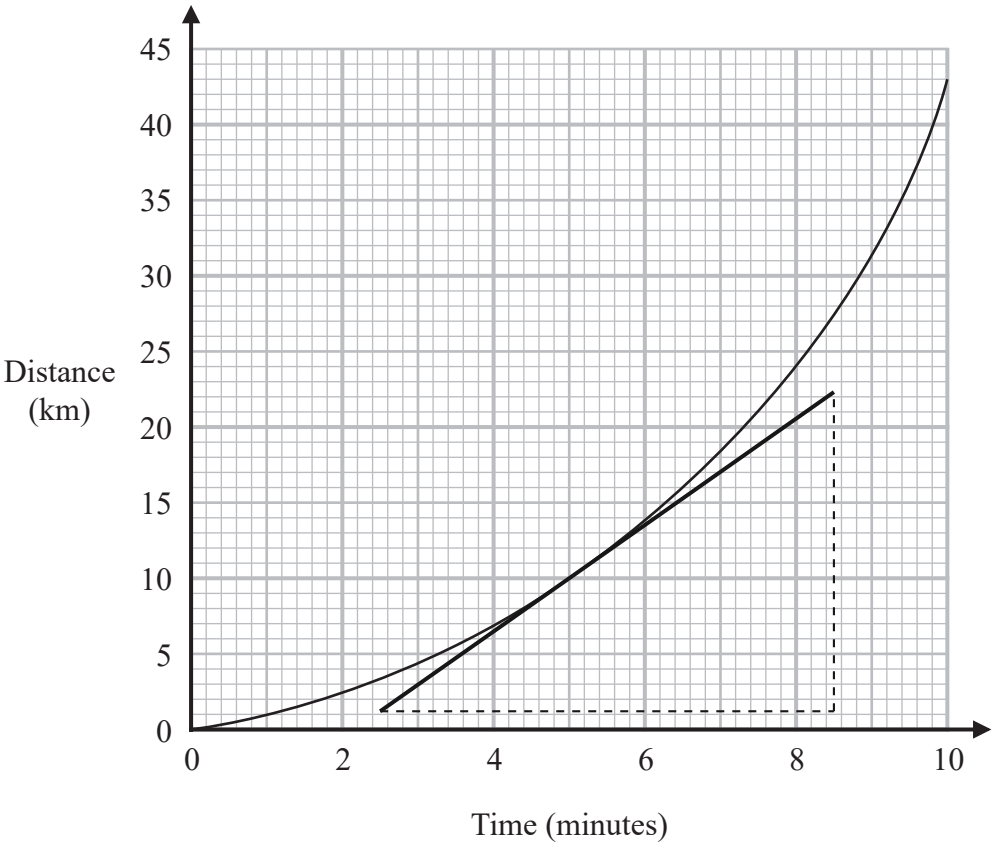
19 Here is a distance-time graph.



(a) Calculate an estimate of the gradient of the graph at time 5 minutes.
You must show how you get your answer.

Draw a tangent touching the actual curve at t = 5.

1 mark
Use a large triangle on the tangent.

Tangent rise ≈ 21.1 km; run = 8.5 – 2.5 = 6 min

1 mark
Gradient = change in distance ÷ change in time.

Gradient ≈ 21.1 ÷ 6 = 3.52 ≈ 3.5 km/min

1 mark
This is an estimate from the tangent.

(3)

(b) Describe what your answer to part (a) represents.

The speed at time 5 minutes.

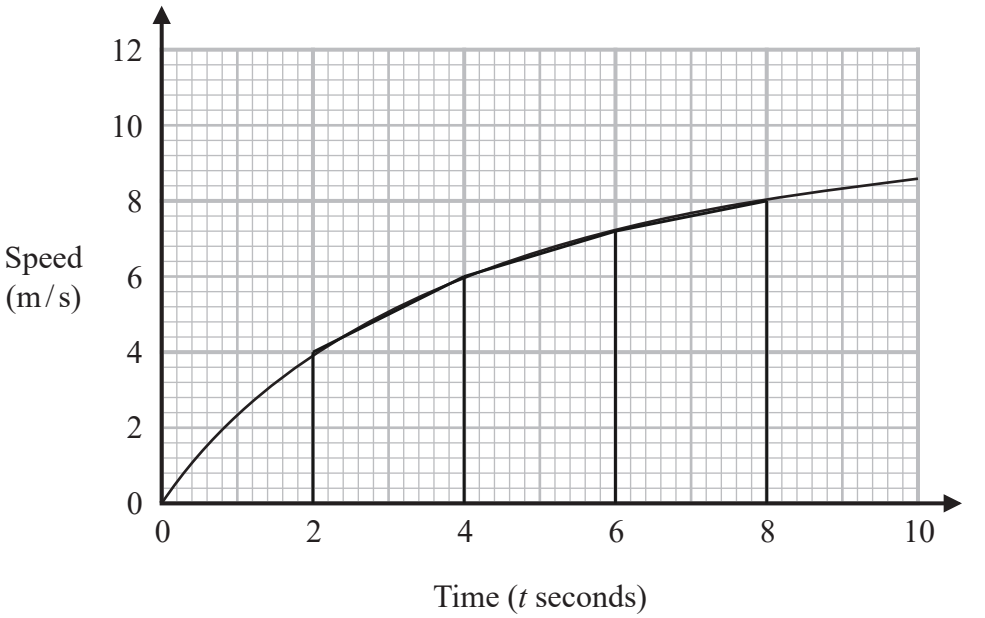
(1)

1 mark
Distance-time gradient represents instantaneous speed.

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Here is a speed-time graph.



(c) Work out an estimate for the area under the graph between $t = 2$ and $t = 8$
Use three strips of equal width.

Three strips from 2 to 8: h = (8 − 2) ÷ 3 = 2 s

Speeds at t = 2, 4, 6, 8 are about 4, 6, 7.2, 8 m/s.

Trapezium area = ½ × (parallel sides) × width

1 mark
First strip: ½ × (4 + 6) × 2 = 10.

Total ≈ ½ × (4 + 6) × 2 + ½ × (6 + 7.2) × 2

+ ½ × (7.2 + 8) × 2

1 mark
Add all THREE trapezia of equal width.

= 10 + 13.2 + 15.2 = 38.4 m

38.4

1 mark
The area estimates the distance travelled from t = 2 to t = 8.

(3)

(Total for Question 19 is 7 marks)